

卢文芳

中山大学“百人计划”副教授、福建省级高层次人才 C 类

美国特拉华大学海洋学、厦门大学物理海洋学双博士

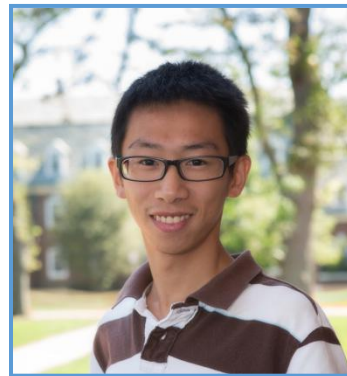
出生年月：1989 年 3 月 籍贯：福建泉州

个人页面：<https://scenty.github.io/personal/>

学院网站：<http://marine.sysu.edu.cn/teacher/8890>

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Research Interests 研究兴趣

海洋动力-生态系统耦合模型，物理海洋学，人工智能海洋学，“深海遥感”和应用，海洋与全球变化

Experience 教育和工作经历

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|---------------|-------------|--|
| • 2022 至今 | 中山大学 | 海洋科学学院 海洋动力过程与气候教研室（系）副主任 |
| • 2022 至今 | 南方海洋实验室（珠海） | 深海生命与生态过程创新团队 |
| • 2018 – 2021 | 福州大学 | 空间数据挖掘与信息共享教育部重点实验室 |
| • 2014 - 2017 | 美国特拉华大学 | Oceanography, 获 XMU-UD 双博士学位 , 导师：严晓海 |
| • 2013 - 2017 | 厦门大学博士 | 物理海洋学，期间 CSC 公派留学 1.5 年 , 导师：江毓武 |
| • 2011 - 2013 | 厦门大学硕士 | 环境管理，之后硕博连读转博士项目 |
| • 2007 - 2011 | 厦门大学本科 | 环境科学，双学士：理学学士，经济学学士 |

Projects 科研项目

- | | | |
|--------------------------|---------------|--------|
| • 珠江口海域中华白海豚对海洋动力响应机制研究 | 中央高校基本科研业务费 | 主持 |
| • 海浪—风暴潮数值模式及综合预警报技术 | 科技部重点研发 | 子任务负责人 |
| • 亚中尺度过程对冬季吕宋藻华现象的生态效应研究 | 国家自然科学基金 | 主持 |
| • 模拟亚中尺度过程对冬季吕宋藻华现象的生态效应 | 中国博士后基金项目二等 | 主持 |
| • 福建省近岸有害藻华机理和过程的模拟研究 | 福建省自然科学基金面上项目 | 主持 |
| • 冬季吕宋藻华过程遥感方法再探，合作者：李芊 | 南海所(LORS)开放课题 | 主持 |

Publications 发表文章

截至 2026 年 6 月，发表一作+通讯 SCI 论文共 20 篇，二区以上 14 篇，包括 Top 期刊 Remote Sens. of Environ., JGR: Oceans, STOTEN, IEEE: TGRS, Remote Sensing 和 Eng Appl Comp Fluid.等。

● 一作+通讯成果 (*代表通讯作者; 分区按最新版中科院分区为准)

1. Wang, T., **Lu, W.** (共一), Lee, Z., Yu, X., Shang, S., & Lin, G. (2026). Reconstructing remote sensing reflectance from MODIS-Aqua for enhanced ocean color applications. *GIScience & Remote Sensing*, 63(1), 2681367. <https://doi.org/10.1080/15481603.2026.2681367> (一区 Top, IF=6.9)
2. He, J., **Lu, W.***, Liu, Y., Yang, G., Jo, Y.-H., & Lai, Z. (2026). Accurate Sea Level Anomaly Forecast over North Pacific with Improved AI Training Strategies. *Ocean-Land-Atmosphere Research*, 5, Article 0128. <https://doi.org/10.34133/olar.0128>
3. Wu, Z., **Lu, W.***, Roobaert, A., Song, L., Yan, X. H., & Cai, W. J. (2025). A machine-learning reconstruction of sea surface pCO₂ in the North American Atlantic Coastal Ocean Margin from 1993 to 2021. *Earth System Science Data*. <https://doi.org/10.5194/essd-17-43-2025> (一区 Top, IF=11.2)
4. Ren, H., **Lu, W.***, Xiao, W., Zhu, Q., Xiao, C., & Lai, Z*. (2024). Intraseasonal response of marine planktonic

ecosystem to summertime Madden-Julian Oscillation in the South China Sea: A model study. *Progress in Oceanography*. <https://doi.org/10.1016/j.pocean.2024.103251> (海洋学二区)

5. Yang, G., Wang, Q., Feng, J., He, L., Li, R., **Lu, W.***, Liao, E., & Lai, Z. (2024). Can Three-dimensional Nitrate Structure be Reconstructed from Surface Information with Artificial Intelligence? – A Proof-of-concept Study. *Science of The Total Environment*. <https://doi.org/10.1016/j.scitotenv.2024.171365> (一区 Top, IF=8.2)
6. Liu, Y., **Lu, W.***, Wang, D., Lai, Z., Ying, C., Li, X., Han, Y., Wang, Z., & Dong, C. (2024). Spatiotemporal wave forecast with transformer-based network: A case study for the northwestern Pacific Ocean. *Ocean Modelling*, 102323. <https://doi.org/10.1016/j.ocemod.2024.102323> (海洋学二区)
7. Wang, T., Sun, Y., Su, H., & **Lu, W.*** (2022). Declined trends of chlorophyll-a in the South China Sea over 2005-2019 from remote sensing reconstruction. *Acta Oceanologica Sinica*. <https://doi.org/10.1007/s13131-022-2097-y> (二区)
8. **Lu, W.**, Gao, X., Wu, Z., Wang, T., Lin, S., Xiao, C., & Lai, Z. (2022). Framework to Extract Extreme Phytoplankton Bloom Events with Remote Sensing Datasets: A Case Study. *Remote Sensing*, 14(15). <https://doi.org/10.3390/rs14153557> (二区 Top, IF=5.349)
9. **Lu, W.**, Wang, J., Jiang, Y., Chen, Z., Wu, W., Yang, L., & Liu, Y. (2022). Data-Driven Method with Numerical Model: A Combining Framework for Predicting Subtropical River Plumes. *Journal of Geophysical Research: Oceans*, e2021JC017925. <https://doi.org/10.1029/2021JC017925> (二区 Top, IF=3.4)
10. Yu, P., W. Xu, X. Zhong, J. A. Johannessen, X.-H. Yan, X. Geng, Y. He, & **Lu, W.*** (2022). A Neural Network Method for Retrieving Sea Surface Wind Speed for C-Band SAR, *Remote Sensing*, 14(9), <https://doi.org/10.3390/rs14092269> (二区 Top, IF= 5.349)
11. Su, H., Zhang, T., Lin, M., **Lu, W.***, & Yan, X.-H. (2021). Predicting subsurface thermohaline structure from remote sensing data based on long short-term memory neural networks. *Remote Sensing of Environment*, 260. <https://doi.org/10.1016/j.rse.2021.112465> (一区 Top, IF=9.085)
12. **Lu, W.**, Su, H., Yang, X., & Yan, X.-H. (2019). Subsurface temperature estimation from remote sensing data using a clustering-neural network method. *Remote Sensing of Environment*, 229, 213-222. <https://doi.org/10.1016/j.rse.2019.04.009> (一区 Top, IF=9.085)
13. **Lu, W.**, Yan, X.-H., & Jiang, Y. (2015). Winter bloom and associated upwelling northwest of the Luzon Island: A coupled physical-biological modeling approach. *Journal of Geophysical Research: Oceans*, 120(1), 533-546. <https://doi.org/10.1002/2014JC010218> (二区 Top, IF=3.559)
14. **Lu, W.**, Jiang, Y., & Lin, J. (2013). Modeling Propagation of 2011 Honshu Tsunami. *Engineering Applications of Computational Fluid Mechanics*, 7(4), 507-518. <https://doi.org/10.1080/19942060.2013.11015489> (一区 Top, IF=5.800)
15. Wang, T., Yu, P., Wu, Z., **Lu, W.***, Liu, X., Li, Q. P., & Huang, B. (2021). Revisiting the Intraseasonal Variability of Chlorophyll-a in the Adjacent Luzon Strait With a New Gap-Filled Remote Sensing Data Set. *IEEE Transactions on Geoscience and Remote Sensing*, 60, 1-11. <https://doi.org/10.1109/tgrs.2021.3067646> (二区 Top, IF=5.858)
16. Ding, Y., Chen, Z., **Lu, W.***, & Wang, X.* (2021). A CatBoost approach with wavelet decomposition to improve satellite-derived high-resolution PM2.5 estimates in Beijing-Tianjin-Hebei. *Atmospheric Environment*, 249. <https://doi.org/10.1016/j.atmosenv.2021.118212> (三区, IF=4.039)
17. **Lu, W.**, Luo, Y.-W., Yan, X., & Jiang, Y. (2018a). Modeling the Contribution of the Microbial Carbon Pump to Carbon Sequestration in the South China Sea. *Science China Earth Sciences*(61), 1594–1604. <https://doi.org/10.1007/s11430-017-9180-y> (二区, IF=3.242)
18. **Lu, W.**, Oey, L.-Y., Liao, E., Zhuang, W., Yan, X.-H., & Jiang, Y. (2018b). Physical modulation to the biological productivity in the summer Vietnam upwelling system. *Ocean Science*, 14(5), 1303-1320.

<https://doi.org/10.5194/os-14-1303-2018> (三区)

19. Su, H., Zhang, H., Geng, X., Qin, T., **Lu, W.***, & Yan, X.-H. (2020). OPEN: A New Estimation of Global Ocean Heat Content for Upper 2000 Meters from Remote Sensing Data. *Remote Sensing*, 12(14), 2294. <https://www.mdpi.com/2072-4292/12/14/2294> (二区, IF=4.509)
20. **Lu, W.**, Yan, X.-H., Han, L., & Jiang, Y. (2017). One-dimensional ocean model with three types of vertical velocities: a case study in the South China Sea. *Ocean Dynamics*, 67(2), 253–262. <https://doi.org/10.1007/s10236-016-1029-9> (四区)

林少文, 任姮烨, **卢文芳***, 2022. 越南沿岸上升流海表叶绿素的季节内尺度变异及机制. *热带海洋学报*.

高心雨, 王天浩, 苏华, 吴文挺, **卢文芳***, 2022. 南海两个代表性海区藻华事件特征的比较研究. *海洋学报*.

卢文芳, 罗亚威, 严晓海, 江毓武, 2018. 模拟“微型生物碳泵”对南海储碳的贡献. *中国科学: 地球科学*.

刘勇, **卢文芳***, 应超, 等. 2019. 日本 311 海啸引起的乐清湾共振研究. *海洋学研究*.

● 学术专著

Lu, W., & Su, H. (2023). Ocean Heat Content Retrieval from Remote Sensing Data Based on Machine Learning. In *Artificial Intelligence Oceanography* (pp. 125-145). Singapore: Springer Nature Singapore.

Su, H., **Lu, W.**, Wang, A., & Zhang, T. (2023). AI-Based Subsurface Thermohaline Structure Retrieval from Remote Sensing Observations. In *Artificial Intelligence Oceanography* (pp. 105-123). Singapore: Springer Nature Singapore.

● 其他成果

Su, H., X. Wu, **W. Lu, W.** Zhang, and X.-H. Yan, 2017: Inconsistent Subsurface and Deeper Ocean Warming Signals During Recent Global Warming and Hiatus. *J. Geophys. Res. Oceans*.

Liao, E., **W. Lu**, X.-H. Yan, Y. Jiang, and A. Kidwell, 2015: The coastal ocean response to the global warming acceleration and hiatus. *Scientific Reports*.

Su, H., Yang, X., **Lu, W.**, et al. 2019. Estimating Subsurface Thermohaline Structure of the Global Ocean Using Surface Remote Sensing Observations. *Remote Sensing*.

He, W., Lai, Z., **Lu, W.**, Li, D., & Chen, C. The effects of atmospheric nitrogen deposition on the marine ecosystem of the northern South China Sea. *Journal of Geophysical Research: Oceans*, e2022JC018790.

Su, H., Wei, Y., **Lu, W.**, Yan, X. H., & Zhang, H. 2023. Unabated Global Ocean Warming Revealed by Ocean Heat Content from Remote Sensing Reconstruction. *Remote Sensing*, 15(3), 566.

Academic Experience 学术经历

审稿记录

- 2025: Limnology and Oceanography: Methods, Journal of Operational Oceanography, Optic Express, RS, 大气科学进展
- 2024 (21): RSE, Ocean-Land-Atmospheric Research, Dynamic Atmosphere and Oceans (2), Limnology and Oceanography, JGR-Oceans, IEEE: Journal of Ocean Engineering, Progress in Oceanography, AOS (3), RS (3), STOTEN (3), TGRS (1), Marine Environmental Research (3), 海洋与湖沼, 集美大学学报
- 2023 (14): Advances in Atmospheric Sciences, Geoscience and Remote Sensing Letter, STOTEN, AOS, JGR-Oceans, RS (3), TGRS, Atmosphere, Frontiers in Marine Science (3), 热带地理
- 2022 (23): Theoretical and Applied Climatology, Journal of Oceanology and Limnology, Frontiers in MS, Journal of Environment Management, Remote Sensing (2), Journal of Marine Science and Engineering (2), IJRS, IEEE: TGRS (3), Applied Science (2), Acta Oceanologica Sinica (2), JGR-Oceans (2), Journal of Marine System, Atmosphere (3), 遥感科学与技术
- 2021 (18): JGR-Oceans, Acta Oceanologica Sinica, IEEE: TGRS, IEEE: JSTARS (4), Remote Sensing (2), International Journal of Remote Sensing (3), International Journal of Environmental Research and Public Health

(2), Sensors (2), Water (2)

- 2020 and earlier (9): GRL, JGR: Oceans (2), Acta Oceanologica Sinica (3), Ocean Science, 海洋科学进展
期刊编辑

- **Youth Editor of *Ocean-Land-Atmosphere Research (OLAR)*, Dec 2023 – present.**

- Guest Editor of *Remote Sensing: Ocean Monitoring Based on Artificial Intelligence and Remote Sensing*, 2023
学术活动

- 口头报告, XMAS-10: Response of marine planktonic ecosystem to subseasonal drivers in the summer South China Sea: A model study, 厦门, 2025 年 1 月
- 邀请报告, MEL: 海洋智能重建及预报, 厦门, 2024 年 10 月
- 口头报告, 第十一届南海年会: Response of marine planktonic ecosystem to summertime subseasonal drivers in the South China Sea: A model study, 大连, 2024 年 11 月
- 邀请报告, 厦门大学 MEL 国重室: 海洋智能重建及预报, 厦门, 2024 年 10 月
- 口头报告, 第二届海岸带大会暨智能海岸论坛: 数值模拟和观测数据结合的机器学习方法实现珠江口水质预测, 北京, 2024 年 9 月
- 口头报告, 21th Annual Meeting of AOGS: Intraseasonal response of marine planktonic ecosystem to summertime Madden-Julian Oscillation in the South China Sea: A model study, PyongChang, Korea, 2024 年 6 月
- 特邀报告, 第三届人工智能海洋学会年会: 人工智能营养盐三维结构重建, 青岛, 2024 年 5 月
- 口头报告, 第二届海洋热含量和海平面青年研讨会, 人工智能海洋热含量重建进展, 青岛, 2024 年 4 月
- Session Co-chair (w/ Dr. Deanesh RAMSEWAK from The University of Trinidad and Tobago), PS-06: DTO Applications for Nearshore and Coastal Systems, 厦门, 2023 年 11 月
- 口头报告, 10th South China Sea Annual Meeting 2023: The planktonic ecosystem's intraseasonal response to summertime Madden-Julian Oscillation in the South China Sea: A model study, 马来西亚吉隆坡, 2023 年 11 月
- 邀请报告/召集人, 第十届海峡两岸海洋环境监测及预报技术: 基于注意力神经网络的西北太平洋波浪预报, 邀请人: 江毓武, 厦门/基隆, 2023 年 11 月
- 口头报告, International Conference on Frontiers of Ocean Science and Technology: Reconstruction of subsurface nitrate concentration in the Indian Ocean with artificial intelligence, 长沙, 2023 年 10 月 (**Best presentation** awarded)
- 邀请报告, 广东海洋大学人工智能海洋学”研究生暑期学校: 人工智能海洋学及其深海遥感应用, 邀请人: 徐广珺, 湛江, 2023 年 7 月
- 特邀报告, 第二届人工智能海洋学论坛: “深海遥感”与海洋数据智能重建: An overview, 珠海, 2023 年 3 月
- 邀请报告, Effects of Tracer Metals on Biological Carbon Pump 研讨会: Intra-seasonal modulation of carbon pump in South China Sea: a model study, 邀请人: 张琼, 香港, 2023 年 3 月
- 邀请报告, 海洋生物地球化学循环研讨会: 生态动力耦合模型研究南海生物碳泵, 邀请人: 朱学明, 线上, 2022 年 12 月
- 邀请报告, 海洋生物泵过程观测与模拟研讨会暨 NSFC 重点项目启动会: 南海生物泵的模拟, 邀请人: 黄邦钦、肖武鹏, 线上, 2022 年 5 月
- 邀请报告, 南海所热带海洋环境国家重点实验室: 人工智能海洋学及其深海遥感应用, 邀请人: 董园, 线上, 2022 年 4 月
- 口头报告, 南海北部跨陆架物质输运的亚中尺度驱动因子, 湛江, 2021
- 邀请报告, 海洋垂直结构遥感探测与反演专题研讨会, 邀请人: 陈戈, 线上, 2020

- 口头报告, The 11th International Workshop on Modeling the Ocean, 无锡, 2019
- 口头报告, 凌峰论坛第 69 讲: 中尺度涡的物理-生物地球化学研究进展与展望, 厦门, 2018
- 口头报告, The first Xiamen Spring School on Ocean Dynamics (XMOD-I), 厦门, 2018
- 口头报告, The third Xiamen Symposium on Marine Environmental Sciences (XMAS-III), 厦门, 2017

其他

- 特拉华学习期间, GPA 3.957/4.0, 选修的 23 个学分, 其中仅有一门为 A-, 其余全 A
- 建立并维持良好的国际 (普林斯顿大学 Leo Oey、特拉华大学) 和国内合作关系;
- 知识问答网站“知乎”海洋/海洋科学话题优秀回答者, “潜入蔚蓝”海洋科普圆桌嘉宾, 多次编辑精选和知乎日报推荐;

Honors and Awards 获奖情况

AOGS2016 海洋学分会最佳海报奖(从 170 余张海报中选出的四张之一), 国际学术会议 ICFOST Best presentation awarded. 博士研究生国家奖学金; 厦门大学校、院、系三级各种奖励多项; 近海海洋环境科学国家重点实验室优秀口头汇报奖等